

Algebraic Topology: a beginner's course

This lecture:

**AlgTopReview3: More on
commutative groups**

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AlgTopReview3: More on commutative groups ①

Structure thm: Every finite commutative (or abelian) group is isomorphic to a direct sum of cyclic groups.

$$\cong \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \oplus \dots \oplus \mathbb{Z}_{n_k}$$

The elements of R.S. are k -tuples of form $(a_1, \dots, a_k)$ where $a_i \in \mathbb{Z}_{n_i}$.

$$n = n_1 n_2 \dots n_k$$

n?

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$$G \cong \mathbb{Z}_{n_1} \oplus \mathbb{Z}_{n_2} \oplus \dots \oplus \mathbb{Z}_{n_k}$$

The elements of R.S. are k -tuples of form $(a_1, a_2, \dots, a_k)$ where $a_i \in \mathbb{Z}_{n_i}$.

Note $|\mathbb{Z}_{n_1} \oplus \dots \oplus \mathbb{Z}_{n_k}| = n_1 n_2 \dots n_k$

Q. What does $\cong$ mean?

Isomorphism ("equality") of groups ②

Def. Groups G, H are isomorphic, written

$G \cong H \iff$ we can find a bijection $\varphi: G \rightarrow H$

satisfying

i) $\varphi(e_G) = e_H$

ii) $\varphi(g_1 * g_2) = \varphi(g_1) * \varphi(g_2)$

iii) $\varphi(g^{-1}) = (\varphi(g))^{-1}$.

For comm. gps, written additively,

i) $\varphi(0_G) = 0_H$

ii) $\varphi(g_1 + g_2) = \varphi(g_1) + \varphi(g_2)$

iii) $\varphi(-a) = -\varphi(a)$.

Note that $\mathbb{Z}_4 \neq \mathbb{Z}_2 \oplus \mathbb{Z}_2$. Why?

④

Because $\mathbb{Z}_4$ contains an element of order 4 but $\mathbb{Z}_2 \oplus \mathbb{Z}_2$ does not.

Def. The order of an element $g \in (G, *)$ is the least positive integer n satisfying

$$\underbrace{g * g * \dots * g}_n = e \quad \boxed{\text{ord}(g) = n}$$

Ex. In $\mathbb{Z}_4$, $\text{ord}(0) = 1$, $\text{ord}(1) = \text{ord}(3) = 4$,
 $\text{ord}(2) = 2$ ♦

Ex. In $\mathbb{Z}_2 \oplus \mathbb{Z}_2$, $\text{ord}((0,0)) = 1$, $\text{ord}((1,0)) = \text{ord}((0,1)) = \text{ord}((1,1)) = 2$ ♦

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Automorphisms of a group G

⑤

Def. An isomorphism $\varphi: G \rightarrow G$ is an automorphism of G .

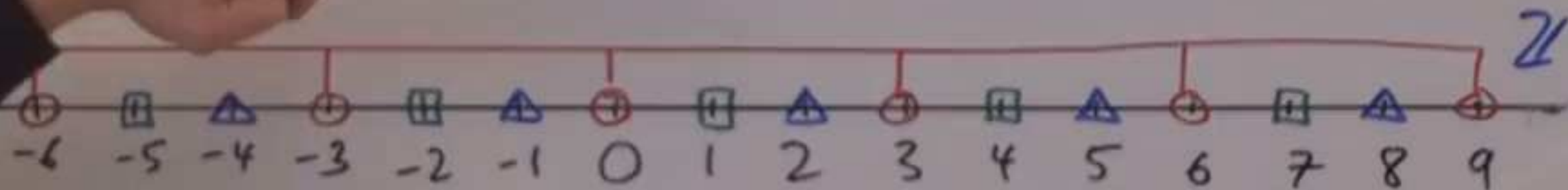
Ex. If $K = \{1, a, b, c\}$ is the Klein 4-gp then a, b, c appear symmetrically in the mult'n table. Hence every permutation of $\{a, b, c\}$ is an automorphism. $\text{Aut}(K) \simeq S_3$ symmetric gp. ♦

Exer. Show that $\text{Aut}(G)$ is a (possibly non-commutative) group for any group G . ♦

Exer. Describe explicitly $\text{Aut}(\mathbb{Z}_n)$ for $n=6, 7, 8$. ♦

subgroups & quotient groups

(7)



$$0: H < \mathbb{Z} \quad 3\mathbb{Z} \equiv H$$

$$\square: 1+H$$

$$\triangle: 2+H$$

~ cosets of H

$$\mathbb{Z}/H = \{H, 1+H, 2+H\}$$

is itself a group:

a quotient group.

$$\mathbb{Z}/H \simeq \mathbb{Z}_3 \text{ (addition(mod 3))}$$

+	H	1+H	2+H
H	H	1+H	2+H
1+H	1+H	2+H	H
2+H	2+H	H	1+H

Ex. $\mathbb{Z}_{12} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$

⑧

$$H = \{0, 3, 6, 9\} \simeq \mathbb{Z}_4 \quad H < \mathbb{Z}_{12}$$

Cosets: $H, 1+H, 2+H$ $\mathbb{Z}_{12}/H \simeq \mathbb{Z}_3$ ♦

Ex. $L = \{0, 4, 8\} \simeq \mathbb{Z}_3 < \mathbb{Z}_{12}$

Cosets: $L, 1+L, 2+L, 3+L$ $\mathbb{Z}_{12}/L \simeq \mathbb{Z}_4$ ♦

Ex. $M = \{0, 2, 4, 6, 8, 10\} \simeq \mathbb{Z}_6$

Cosets: $M, 1+M$

+	M	1+M
M	M	1+M
1+M	1+M	M

$\mathbb{Z}_{12}/M \simeq \mathbb{Z}_2$ ♦

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also see:

**AlgTopReview4: Free groups
and non-commutative groups**